A Business-Oriented Virality Framework for Evaluating Instagram Reel Performance Using Business Virality Index (BVI)

Section 1: Introduction

1.1) Business Context

Instagram Reels has emerged as one of the fastest-growing short-form video platforms, with millions of creators publishing content every day. However, the distribution of content performance is highly uneven. While a small percentage of reels achieve exceptional reach and engagement through viral distribution, the majority fail to extend beyond their creators' existing follower base.

This variation suggests that virality is influenced by a combination of content quality, audience engagement, and platform-driven distribution signals rather than follower count alone. Consequently, understanding the characteristics that differentiate high-performing reels from low-performing ones has become a critical business objective for Instagram's Creator Insights team.

Several platform signals are believed to influence reel performance, including:

- Content quality and hook effectiveness
- Use of trending audio
- Viewer retention and watch behaviour
- Engagement through likes, comments, shares, and saves
- Algorithmic amplification through Explore Page recommendations
- Reach among non-followers

Understanding the relationship between these factors enables Instagram to improve creator recommendations, optimize content distribution strategies, and enhance overall user engagement on the platform.

1.2) Business Problem Statement

Although Instagram provides an **Original Virality Score** as a composite indicator of reel performance, it remains unclear whether this metric adequately represents business success.

Creators with similar follower counts frequently achieve significantly different levels of reach, engagement, and organic distribution. Furthermore, the existing virality score does not explicitly communicate whether it effectively captures the business outcomes that matter most, such as expanding non-follower reach, increasing audience engagement, and maximizing content distribution.

As a result, Instagram's Creator Insights team requires a comprehensive analytical framework to evaluate whether the existing virality score accurately reflects business performance and, if necessary, develop a more business-oriented performance metric.

1.3) Project Objectives

The primary objective of this study is to identify the content, behavioural, and distribution factors that contribute to the business success of Instagram Reels.

Specifically, the study aims to:

- Evaluate whether the existing Original Virality Score effectively represents business performance.
- Develop a Business Virality Index (BVI) that better reflects business-oriented reel performance.
- Segment reels into meaningful business performance categories.
- Identify the characteristics that distinguish high-performing reels from low-performing reels.
- Generate actionable recommendations that can assist creators in improving content performance and help Instagram optimize its recommendation algorithms.

1.4) Scope of the Study

This study analyses **100,000 Instagram Reels** to understand the drivers of business virality. The analysis focuses on evaluating the effectiveness of the existing Original Virality Score, constructing a Business Virality Index (BVI), performing exploratory data analysis, identifying performance patterns across virality categories, and presenting findings through an interactive Tableau dashboard.

The study does not attempt to model Instagram's proprietary recommendation algorithm. Instead, it uses the available dataset to derive data-driven business insights and practical recommendations for creators and platform stakeholders.

Section 2: Exploratory Data Analysis (EDA)

2.1) Dataset Overview

Before evaluating the effectiveness of the Original Virality Score, an exploratory data analysis (EDA) was performed to assess the quality, completeness, and characteristics of the Instagram Reels dataset. This step ensured that the dataset was suitable for subsequent business analysis and metric development.

The dataset consists of 100,000 Instagram Reels collected across 100,000 unique creators, with each record representing a unique reel. The dataset contains 24 variables describing creator characteristics, content quality, audience engagement, viewer behaviour, and algorithmic distribution signals.

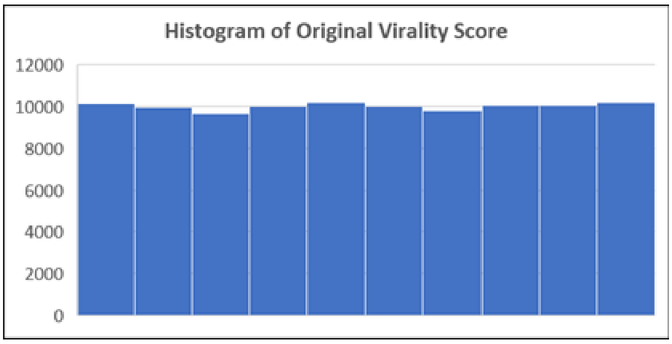
Key EDA Findings

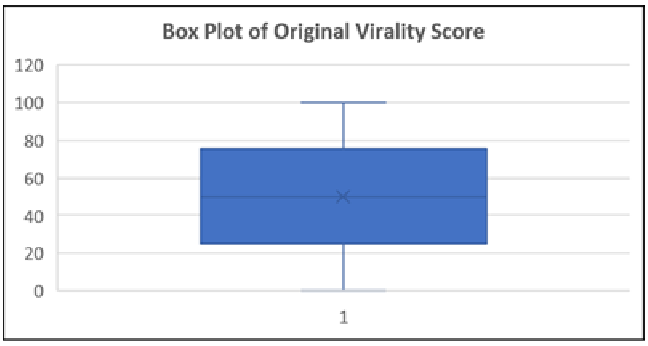
- Total Records: **100,000 Instagram Reels**
- Unique Reel IDs: **100,000**
- Unique Creator IDs: **100,000**
- Total Variables: **24**
- Missing Values: **None (0% missing across all variables)**
- Duplicate Records: **None**

The absence of missing values and duplicate records indicates that the dataset is complete and suitable for statistical analysis without requiring additional data cleaning or imputation. Furthermore, the dataset contains variables representing multiple dimensions of reel performance, including content quality, viewer retention, audience engagement, creator influence, and algorithmic distribution, providing a comprehensive foundation for evaluating business virality.

2.2) Distribution of the Original Virality Score

Before evaluating the effectiveness of the Original Virality Score, it is important to understand its statistical distribution across the dataset. Examining the score distribution helps determine whether the metric is balanced and suitable for comparative analysis.



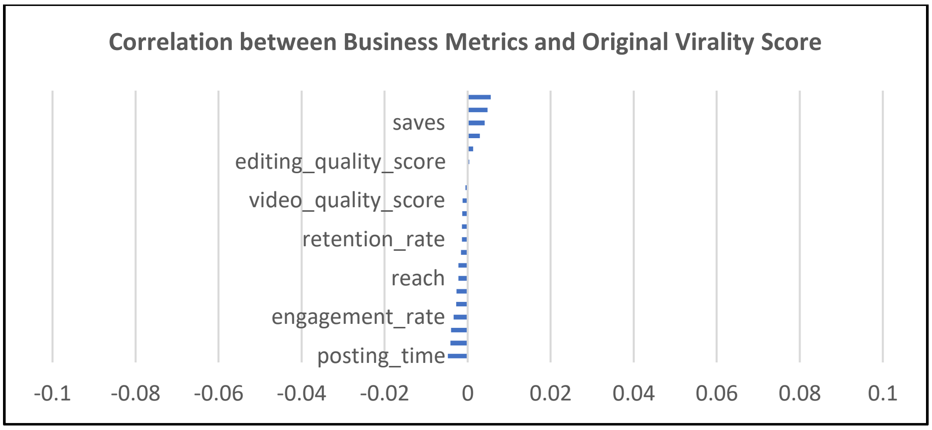


- Scores are evenly distributed across the 0–100 range.
- No significant skewness is observed.
- The box plot indicates the absence of major outliers.
- The distribution is suitable for comparative analysis.

Although the score is statistically well distributed, a balanced distribution alone does not confirm that it accurately represents business performance. Further analysis is required to assess whether it aligns with meaningful business outcomes.

2.3) Correlation Analysis of the Original Virality Score

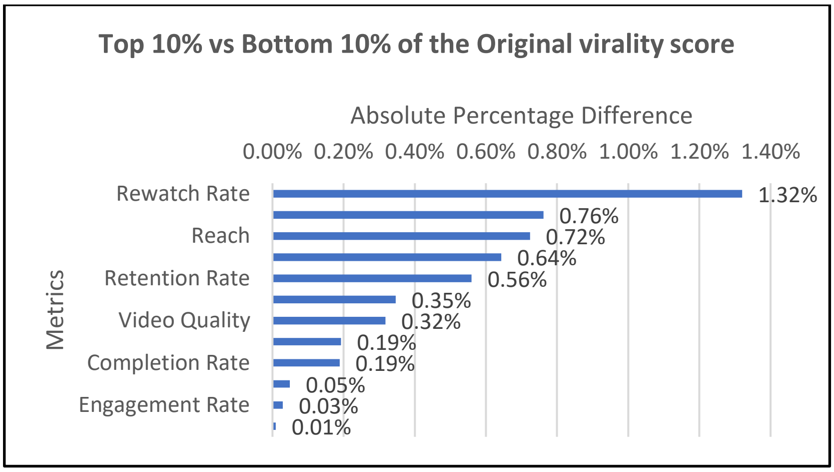
To determine whether the Original Virality Score is an effective indicator of business success, its relationship with key business performance metrics was examined. A strong correlation would suggest that the existing score adequately represents reel performance, whereas weak relationships would indicate the need for a more business-oriented metric.



- All business metrics exhibit negligible correlations with the Original Virality Score, with correlation coefficients remaining close to zero.
- Metrics traditionally associated with content success, including Reach, Engagement Rate, Shares, Saves, and Non-Follower Reach Ratio, do not demonstrate meaningful relationships with the existing score.
- These findings suggest that the Original Virality Score does not adequately capture the business outcomes that define successful Instagram Reels, motivating the development of the Business Virality Index (BVI).

2.4) Comparative Performance Analysis of Top and Bottom Reels

To further assess the effectiveness of the Original Virality Score, the top 10% and bottom 10% of reels were compared across key business performance metrics. If the existing virality score accurately reflects business success, the highest-scoring reels would be expected to demonstrate substantially better performance across engagement, distribution, and content quality metrics.



- The differences between the top and bottom 10% reels are relatively small across all business metrics.
- Rewatch Rate (1.32%) shows the highest variation, followed by Non-Follower Reach Ratio (0.76%) and Reach (0.72%).
- Core engagement metrics such as Engagement Rate (0.03%) and Shares (0.01%) exhibit almost no difference between the two groups.
- Content quality indicators, including Video Quality, Editing Quality, and Hook Strength, also show only marginal improvements.
- The limited separation between the highest- and lowest-scoring reels suggests that the Original Virality Score does not effectively distinguish strong business performance.

An effective business performance metric should clearly differentiate high-performing reels from low-performing ones across key business outcomes. However, the Original Virality Score shows only marginal differences between the top and bottom performers for most business metrics. This indicates that the existing score does not adequately capture the factors that drive meaningful business success, reinforcing the need for a more comprehensive and business-oriented performance metric.

2.5) Hypothesis Validation of the Original Virality Score

To validate whether the Original Virality Score accurately represents business success, six business hypotheses were formulated based on the key mechanisms believed to influence Instagram Reel performance. These hypotheses examined whether higher virality scores consistently reflected improvements in content quality, audio strategy, viewer retention, engagement, algorithmic distribution, and creator influence.

Business Metric	Hypothesis Result
Content Quality	Not Supported
Audio Strategy	Not Supported
Viewer Retention	Not Supported
Engagement	Not Supported
Algorithmic Distribution	Not Supported
Creator Influence	Not Supported

- None of the six business hypotheses were supported by the Original Virality Score.
- Higher virality scores did not consistently correspond to stronger content quality, audience engagement, or viewer retention.
- Distribution-related factors, such as Non-Follower Reach and Explore Page Boost, also showed no meaningful relationship with the Original Virality Score.
- Creator size alone did not explain variations in the existing virality score.
- These findings indicate that the Original Virality Score does not reliably represent the business mechanisms that contribute to successful reel performance.

The hypothesis validation confirms the findings from the correlation and comparative analyses. Despite being statistically well distributed, the Original Virality Score fails to capture the content, behavioural, and distribution factors that define business success on Instagram Reels. Consequently, relying solely on the existing metric may lead to inaccurate evaluation of reel performance. These results provide a strong justification for developing a more business-oriented performance metric—the **Business Virality Index (BVI)**.

2.6) Why a Business Virality Index is Needed

After evaluating the Original Virality Score through distribution analysis, correlation analysis, comparative performance analysis, and hypothesis validation, it is evident that the existing metric does not adequately represent business success. The score demonstrates weak relationships with key business outcomes and fails to distinguish high-performing reels based on the mechanisms that drive virality. Therefore, a new metric is required to measure reel performance from a business perspective by incorporating engagement, reach, distribution, and audience behaviour into a unified index.

Section 3: Development of the Business Virality Index (BVI)

3.1) Selection of Business Performance Indicators

Based on the findings from the previous analyses and Instagram's business objectives, a new composite metric named the Business Virality Index (BVI) was developed to better evaluate reel performance. Rather than relying on a single existing score, the BVI combines multiple business-oriented performance indicators that collectively represent audience reach, engagement, and organic distribution.

The five KPIs selected for constructing the Business Virality Index are presented below.

KPI	Business Significance
Non-Follower Reach Ratio	Measures the effectiveness of algorithmic distribution beyond the creator's existing audience.
Shares	Indicates how frequently users distribute the content to others, representing viral potential.
Engagement Rate	Reflects audience interaction relative to content exposure.
Saves	Indicates the audience want to get back to the reel again.
Reach	Measures the number of unique users exposed to the reel, representing audience expansion.
Impressions	Captures the overall visibility of the reel, including repeated views.

- The selected KPIs represent the three key dimensions of business success: **visibility, engagement, and distribution**.
- Each KPI measures a distinct aspect of reel performance, reducing reliance on a single metric.
- Combining multiple business indicators provides a more comprehensive assessment of virality than the Original Virality Score.

The Business Virality Index was designed to evaluate reel performance from a business perspective by integrating complementary performance indicators into a single composite metric. This approach provides a more holistic representation of content success and supports more meaningful comparisons across reels.

3.2) Construction of the Business Virality Index

Based on the evaluation presented in the previous section, a new composite Business Virality Index (BVI) was developed to provide a comprehensive measure of Instagram Reel performance. Since each KPI is measured on a different scale, the selected variables were first normalized using Min–Max Normalization. The normalized values were then combined using weighted aggregation to produce a single Business Virality Index ranging from 0 to 100.

The Original Virality Score was intentionally excluded from the construction of the Business Virality Index. As demonstrated in the previous section, the existing score exhibited weak relationships with key business performance indicators, failed to differentiate high-performing reels from low-performing ones, and did not support any of the proposed business hypotheses. Including the Original Virality Score in the new index would have carried forward the limitations identified during the evaluation. Therefore, the Business Virality Index was developed exclusively using business-oriented performance indicators that directly represent audience reach, engagement, and organic distribution.

Step 1: Min–Max Normalization

To ensure comparability among KPIs with different units and ranges, each selected metric was transformed using Min–Max Normalization.

$$\text{Normalized Value} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

This transformation scales every KPI to a common range of **0 to 1**, preventing metrics with larger numerical values from dominating the composite score.

Step 2: Weighted Aggregation

The normalized KPIs were assigned weights based on their relative importance in representing business-oriented reel performance. Greater emphasis was placed on metrics that directly capture organic distribution and audience-driven content amplification.

Metric	Weight	Business Justification
Non-Follower Reach Ratio	30%	Best indicator of organic algorithmic spread beyond followers
Shares	25%	Direct amplification through user sharing
Engagement Rate	20%	Overall audience interaction normalized by reach
Saves	10%	Indicates long-term content value
Reach	10%	Measures audience scale but may be influenced by creator size
Impressions	5%	Exposure volume; weakest standalone indicator

The weights were assigned based on business relevance and Instagram's stated objective of identifying the factors that drive organic content distribution and creator success. Greater importance was given to Non-Follower Reach Ratio and Shares because they directly reflect algorithmic amplification and user-driven content sharing, while Reach and Impressions were assigned lower weights as supporting visibility metrics.

Step 3: Business Virality Index Formula

The Business Virality Index was calculated using the following weighted composite equation:

BVI=(0.30×NFRR+0.25×Shares+0.20×EngagementRate+0.10×Saves+0.10×Reach+0.05×Impressions) ×100

where each variable represents its normalized value.

3.3) Business Virality Categories

To simplify interpretation and facilitate business decision-making, the Business Virality Index (BVI) was segmented into four performance categories: Low, Medium, High, and Viral. These categories enable meaningful comparison of reel performance and provide an intuitive framework for identifying content with varying levels of business success. Using fixed numerical ranges could have resulted in uneven category sizes due to the underlying distribution of BVI scores. Quartile-based segmentation was therefore adopted to create balanced and representative performance groups.

The quartile-based approach was selected because it creates categories based on the actual distribution of the data rather than arbitrary numerical boundaries.

This approach offers several advantages:

- It ensures that each category represents a meaningful segment of the dataset rather than predetermined score intervals.
- It reduces the impact of skewed or uneven score distributions that may occur when using fixed boundaries.
- It creates balanced performance groups, making comparisons across categories more reliable.
- It enables the classification framework to adapt naturally to the characteristics of the dataset, improving the interpretability of the analysis.

As a result, the Business Virality Categories provide a more representative segmentation of reel performance than fixed score thresholds.

Category	BVI Range	Interpretation
Low	Q1	Lowest 25% of Business Virality Index scores.
Medium	Q2	Average-performing reels with moderate business impact.
High	Q3	Strong-performing reels with above-average business performance.
Viral	Q4	Top 25% of reels demonstrating exceptional business performance.

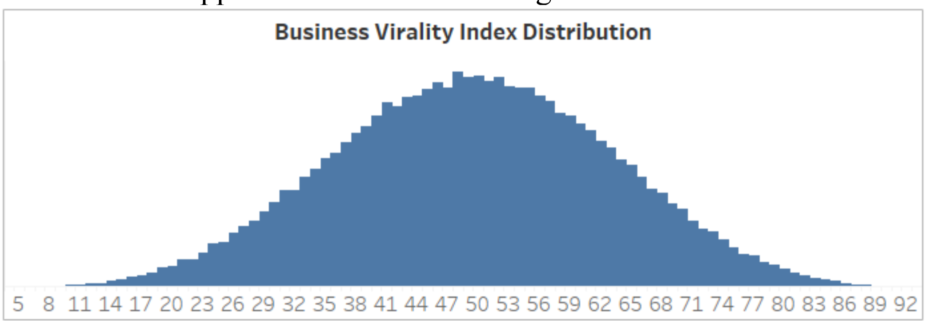
Using quartile-based segmentation ensures that the Business Virality Categories reflect the actual distribution of reel performance within the dataset. This approach provides a robust and data-driven framework for comparing content performance and supports more meaningful business analysis in the subsequent sections.

Section 4: Business Performance Analysis using Business Virality Index

Following the development of the Business Virality Index (BVI), business performance analysis was conducted to evaluate its effectiveness in distinguishing high-performing reels from low-performing ones. Unlike the Original Virality Score, which demonstrated weak relationships with key business metrics, the BVI was designed to capture audience reach, engagement, and organic distribution. The following analyses assess whether the BVI successfully reflects these business performance dimensions.

4.1) Distribution of the Business Virality Index

Before analysing individual business metrics, it is important to examine how the Business Virality Index is distributed across the dataset. A balanced distribution indicates that the index provides meaningful variation in performance and supports reliable business segmentation.

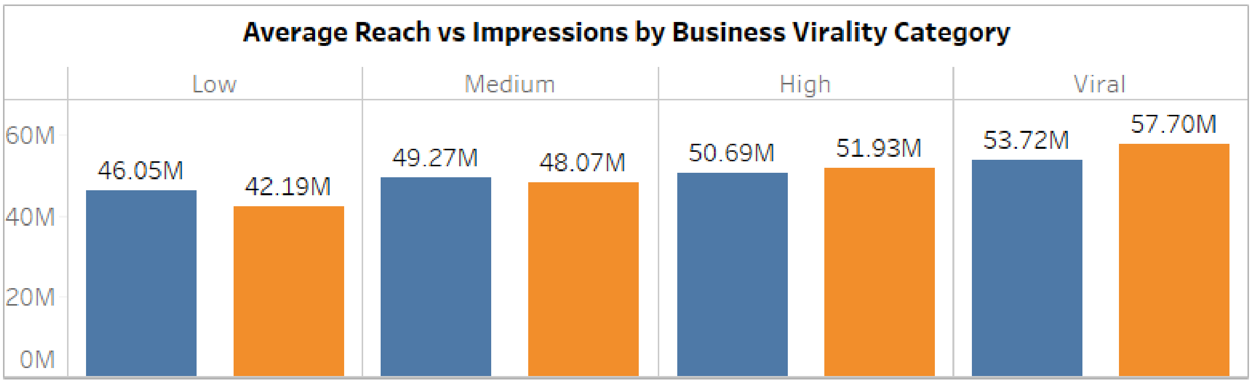


- The Business Virality Index exhibits an approximately normal distribution across the dataset.
- Most reels are concentrated within the Medium and High performance ranges.
- Relatively fewer reels fall into the extreme Low and Viral categories.
- The distribution provides sufficient variation for meaningful comparison across business performance groups.

Having established that the Business Virality Index provides a balanced representation of reel performance, the subsequent analyses evaluate its ability to distinguish key business outcomes across the defined Business Virality Categories.

4.2) Reach and Visibility Analysis

Audience visibility is a critical indicator of content performance on Instagram Reels, reflecting how effectively a reel reaches and is exposed to users. To evaluate whether the Business Virality Index captures this aspect of performance, the average Reach and Impressions were analysed across the four Business Virality Categories.

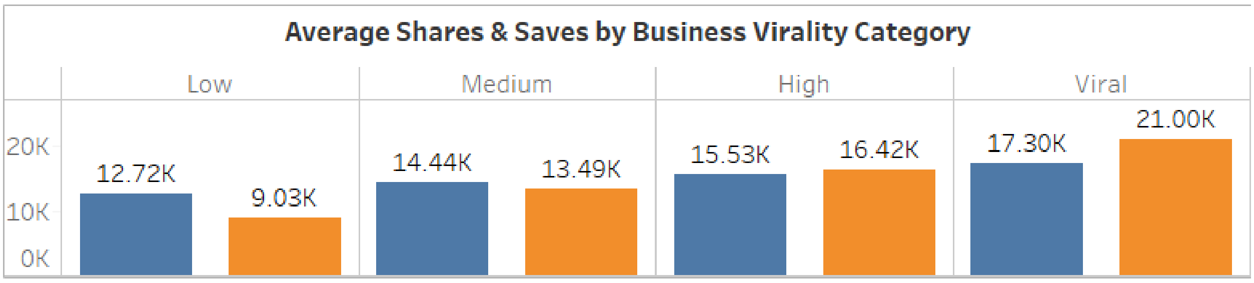


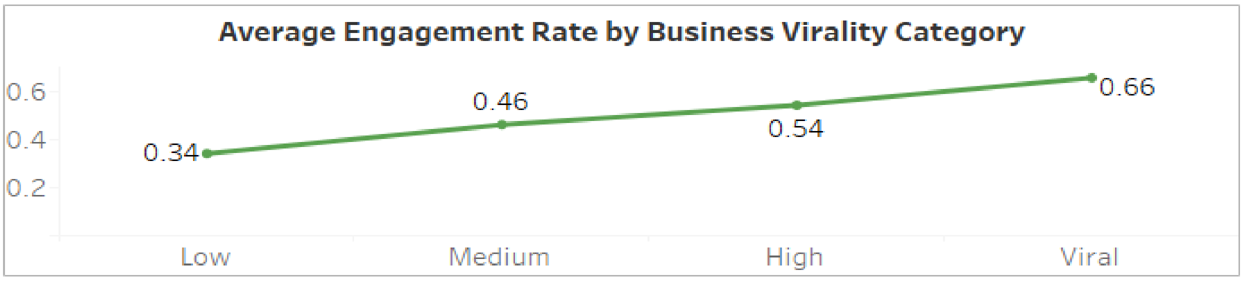
- Both average Reach and average Impressions increase progressively from the Low to Viral Business Virality Categories.
- Viral reels achieve the highest values for both metrics, indicating substantially greater audience visibility than Low-performing reels.
- The consistent upward trend across all four categories demonstrates a clear relationship between increasing BVI and content visibility.
- The separation between categories indicates that the Business Virality Index effectively differentiates reels based on audience exposure.

The observed increase in Reach and Impressions across the Business Virality Categories supports the hypothesis that higher-performing reels achieve greater audience visibility. Unlike the Original Virality Score, which showed weak relationships with business metrics, the Business Virality Index clearly distinguishes reels according to their ability to reach larger audiences, demonstrating that it is a more effective measure of business performance.

4.3) Audience Engagement Analysis

Audience engagement reflects how users interact with Instagram Reels and is a key indicator of content quality and viewer interest. To evaluate whether the Business Virality Index effectively captures audience engagement, the average Shares, Saves, and Engagement Rate were analysed across the four Business Virality Categories.





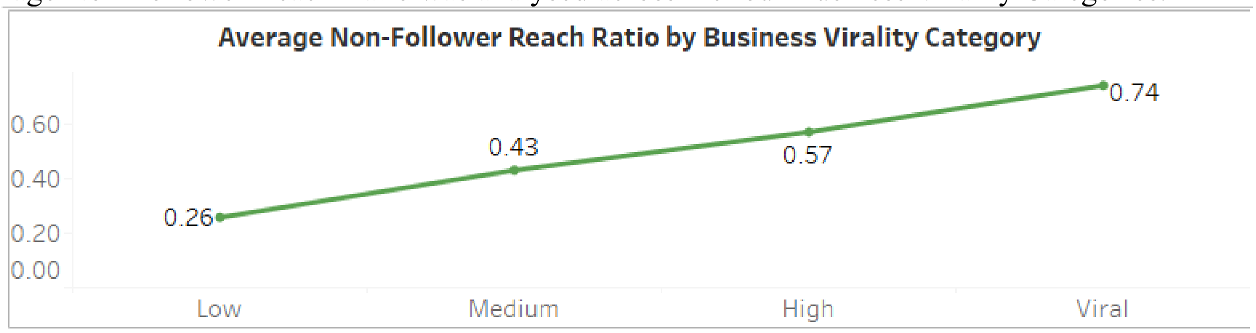
- Average **Shares** and **Saves** increase consistently from the **Low** to **Viral** Business Virality Categories.
- Viral reels generate the highest average Shares and Saves, indicating stronger audience interaction and content dissemination.
- The **Engagement Rate** also shows a steady upward trend across all four categories, increasing from **0.34** for Low-performing reels to **0.66** for Viral reels.
- The consistent improvement across all three-engagement metrics demonstrates that higher BVI scores are associated with stronger audience participation.

The increasing trend in Shares, Saves, and Engagement Rate supports the engagement hypothesis, indicating that reels with higher Business Virality Index scores generate greater audience interaction. Unlike the Original Virality Score, which failed to demonstrate meaningful relationships with engagement metrics, the Business Virality Index effectively differentiates reels based on their ability to encourage user participation and content sharing.

The Business Virality Index clearly distinguishes engagement performance across the four Business Virality Categories. The progressive increase in Shares, Saves, and Engagement Rate demonstrates that reels with higher BVI consistently generate stronger audience interaction, validating the proposed engagement hypothesis.

4.4) Algorithmic Distribution Analysis

One of the primary objectives of Instagram's recommendation system is to distribute high-quality content beyond a creator's existing audience. The Non-Follower Reach Ratio measures the proportion of views generated from users who do not already follow the creator, making it an important indicator of algorithmic amplification. To evaluate whether the Business Virality Index captures this behaviour, the average Non-Follower Reach Ratio was analysed across the four Business Virality Categories.



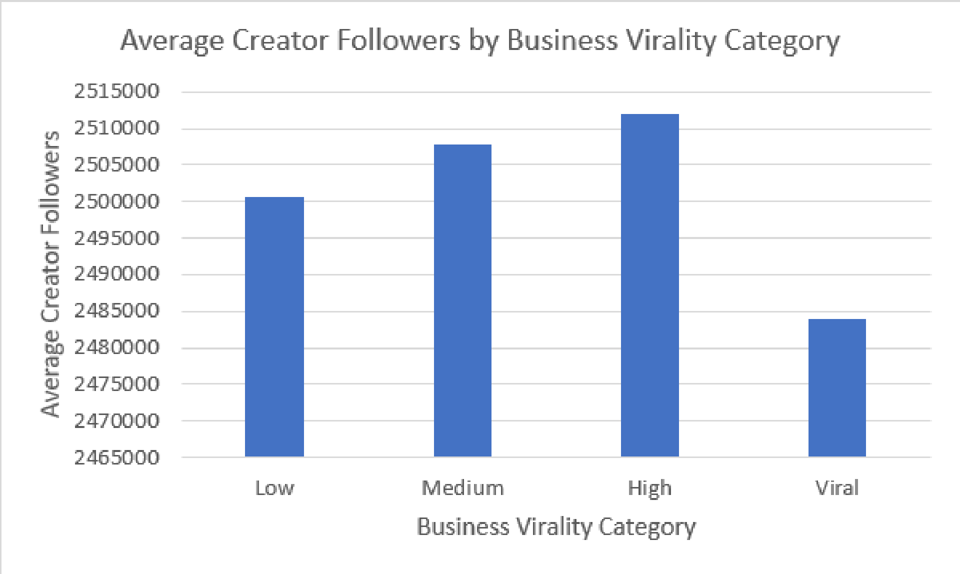
- The Non-Follower Reach Ratio increases consistently across the Business Virality Categories, from 0.26 for Low-performing reels to 0.74 for Viral reels.
- Viral reels receive a substantially higher proportion of views from non-followers compared to Low-performing reels.
- The steady increase across all four categories indicates a strong relationship between Business Virality Index and algorithmic distribution.
- The clear separation between categories demonstrates that the Business Virality Index effectively differentiates reels based on their ability to expand beyond their existing audience.

The progressive increase in the Non-Follower Reach Ratio supports the algorithmic distribution hypothesis, indicating that reels with higher Business Virality Index scores are more likely to be recommended to users outside the creator's follower base. Unlike the Original Virality Score, which did not demonstrate meaningful relationships with distribution metrics, the Business Virality Index effectively captures a reel's organic distribution potential.

The Business Virality Index demonstrates a clear and consistent increase in Non-Follower Reach Ratio across the Business Virality Categories. This indicates that reels with higher BVI scores receive greater exposure beyond their existing audience, validating the role of algorithmic distribution in driving business virality.

4.5) Creator Influence Analysis

Although creators with larger follower bases are often expected to achieve better content performance, Instagram aims to determine whether creator size alone is sufficient to drive business virality. To evaluate this, the average creator follower count was compared across the Business Virality Categories.



- Average creator followers increase slightly from the Low to High categories.
- However, the Viral category has a lower average follower count than the High category.
- Despite having fewer followers, Viral reels achieve the highest Business Virality Index due to stronger engagement and broader non-follower distribution.
- The variation in follower count across categories is relatively small compared to the substantial improvements observed in engagement and algorithmic distribution metrics.

The findings indicate that creator size alone does not determine business virality. While larger audiences may provide greater initial visibility, truly viral reels are driven by content performance, audience engagement, and recommendation system amplification. This aligns with Instagram's objective of enabling high-quality content to reach large audiences regardless of creator size.

Although creator followers increase marginally from the Low to High categories, the Viral category does not have the highest average follower count. This indicates that follower size alone is insufficient to explain business virality. Instead, engagement quality and algorithmic distribution play a much more significant role in driving viral performance.

Section 5: Business Hypothesis Evaluation

5.1) Overview

The Business Virality Index was developed to evaluate reel performance using business-oriented engagement and distribution indicators. The analyses presented in the previous section were used to assess the research hypotheses defined at the beginning of the study. Unlike the Original Virality Score, which showed little relationship with business performance metrics, the Business Virality Index provided clear evidence supporting most of the proposed business mechanisms.

5.2) Hypothesis Validation Summary

Mechanism	Hypothesis	Result	Key Evidence
Content Quality	Higher-quality content increases business virality	Not Supported	Content quality metrics showed no consistent increase across BVI categories.

Audio Strategy	Trending audio improves business virality	Not Supported	Audio popularity varied minimally across categories.
Viewer Retention	Better watch behaviour leads to higher business virality	Not Supported	Retention, Completion, Rewatch Rate, and Watch Time remained nearly constant across all categories.
Engagement	Greater audience engagement improves business virality	Supported	Engagement Rate, Shares, and Saves increased consistently from Low to Viral categories.
Algorithmic Distribution	Greater non-follower exposure drives business virality	Supported	Non-Follower Reach Ratio increased from 0.26 to 0.74 across categories.
Creator Influence	Larger follower base leads to higher business virality	Not Supported	Viral reels did not have the highest average creator follower count.

5.3) Key Findings

The hypothesis evaluation reveals that business virality is primarily driven by **audience engagement** and **algorithmic distribution** rather than creator characteristics or individual content attributes. Reels that generate higher engagement and reach a greater proportion of non-followers consistently achieve higher Business Virality Index scores. Conversely, content quality indicators, audio strategy, viewer retention metrics, and creator follower count do not independently explain business virality within this dataset. These findings suggest that maximizing audience interaction and organic distribution should be the primary focus for improving reel performance.

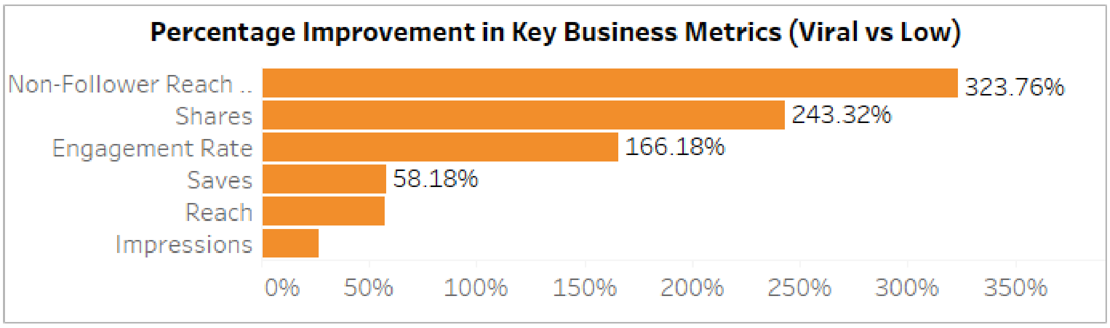
Section 6: Conclusion & Business Recommendations

6.1) Overall Conclusion

This study evaluated whether Instagram's Original Virality Score accurately represents business performance. The findings demonstrated that the original score showed weak relationships with key business metrics and failed to support most of the proposed business hypotheses. To address this limitation, a Business Virality Index (BVI) was developed using weighted engagement and distribution metrics. The BVI provided a more meaningful representation of reel performance and successfully identified audience engagement and algorithmic distribution as the primary drivers of business virality.

6.2) Executive Summary of Key Business Improvements

Percentage increase in average metric values between Viral and Low Business Virality Categories.



- Viral reels achieve over 323% higher Non-Follower Reach, making algorithmic distribution the strongest differentiator.
- Shares and Engagement Rate improve by 243% and 166%, respectively, highlighting audience interaction as a key driver of business virality.
- Reach, Saves, and Impressions improve to a lesser extent, indicating that expanding content beyond existing followers contributes more to virality than increasing overall exposure alone.
- The Original Virality Score does not adequately explain business performance.

- Business Virality Index provides a more business-oriented evaluation framework.
- Audience engagement is one of the strongest indicators of business virality.
- Non-Follower Reach Ratio is the most influential driver of successful reel distribution.
- Creator follower count alone does not determine viral performance.
- Viewer retention, content quality indicators, and audio strategy showed limited influence in this dataset.

6.3) Business Recommendations

The Business Virality Index (BVI) analysis identified the key business factors associated with successful Instagram Reels. Based on the findings from the dashboard analysis and hypothesis evaluation, the following recommendations are proposed for Instagram's Creator Insights team and content creators. Each recommendation is directly supported by evidence obtained during the analysis.

Prioritize Algorithmic Distribution

The analysis identified algorithmic distribution as the strongest driver of business virality. Viral reels achieved **323.76% higher Non-Follower Reach Ratio** than low-performing reels, while the average Non-Follower Reach Ratio increased steadily from **0.26 (Low) to 0.74 (Viral)** across Business Virality Categories. These findings indicate that expanding content beyond a creator's existing audience contributes more to business success than simply increasing impressions.

Instagram should continue optimizing recommendation algorithms to maximize Explore Page exposure, while creators should focus on producing content that encourages discovery among non-followers.

Encourage Shareable Content

Audience sharing behaviour emerged as one of the strongest indicators of business success. Viral reels generated **243.32% higher average shares** than low-performing reels, and average shares increased from **9.03K** in the Low category to **21.00K** in the Viral category.

Creators should design content with strong emotional appeal, educational value, or entertainment potential that naturally encourages viewers to share the reel. Instagram can reinforce this behaviour by recommending content with higher sharing potential.

Improve Meaningful Audience Engagement

Engagement analysis demonstrated a consistent increase in audience interaction across Business Virality Categories. Average Engagement Rate increased from **0.34** for Low-performing reels to **0.66** for Viral reels, while Viral reels achieved a **166.18% improvement** in Engagement Rate compared with Low-performing reels.

Rather than optimizing for likes alone, creators should focus on generating meaningful interactions through comments, saves, and shares, as these signals are more closely associated with business virality.

Focus on Business-Oriented Performance Metrics

The exploratory analysis showed that the **Original Virality Score exhibited negligible correlations with all major business metrics**, making it unsuitable for evaluating business performance. In contrast, the Business Virality Index demonstrated clear and consistent improvements across Reach, Shares, Engagement Rate, and Non-Follower Reach.

Instagram should consider adopting a business-oriented performance metric similar to the Business Virality Index when evaluating creator success and optimizing recommendation strategies.

Creator Size Alone Should Not Drive Recommendations

The Creator Influence analysis showed only minor differences in average follower counts across Business Virality Categories, and the hypothesis that larger creators consistently achieve higher virality was not supported.

Recommendation systems should prioritize content quality and engagement signals rather than relying heavily on creator follower count, ensuring that high-quality content from smaller creators also has opportunities to reach wider audiences.

6.5) Limitations of the Study

- Dataset Scope: - The analysis was conducted using a single Instagram Reels dataset containing 100,000 observations. While the dataset was sufficiently large for exploratory analysis, the findings may not fully represent content performance across different regions, time periods, or user segments.

- Business Virality Index Weight Assignment: - The Business Virality Index (BVI) was developed using manually assigned weights based on business relevance and literature-informed judgment. Although these weights provide a meaningful business perspective, they were not statistically optimized using machine learning or optimization techniques.
- Limited Platform and Algorithm Information: - Instagram's recommendation algorithm is proprietary, and several internal ranking signals were unavailable for analysis. Consequently, the study could evaluate only the observable engagement and distribution metrics present in the dataset.
- Exclusion of Qualitative Content Factors: - The analysis primarily focused on quantitative performance metrics and did not consider qualitative aspects such as storytelling quality, creativity, visual aesthetics, emotional appeal, or caption sentiment. These factors may also influence reel performance but were beyond the scope of the available data.
- Correlation-Based Business Analysis: - The study relied on exploratory data analysis and comparative business analysis to identify relationships between performance metrics and business virality. As a result, the findings indicate associations rather than definitive cause-and-effect relationships.

The above limitations provide opportunities for extending the Business Virality Index framework through more advanced analytical techniques and broader datasets, as discussed in the future scope.

6.5) Future Scope

- Develop a machine learning model to predict Business Virality Index.
- Optimize BVI weights using statistical or AI-based approaches.
- Validate the framework using real-world Instagram data.
- Incorporate additional content features such as sentiment, visual aesthetics, or caption semantics.
- Build a real-time creator performance dashboard for continuous monitoring.